

NADRA FAQ Assistant — Initial Strategic Plan

RAG-based question answering for NADRA services | Phase 1 | July 2026

1. The Problem

Citizens have common, repetitive questions about NADRA services — CNIC, NICOP, POC, CRC (Child Registration Certificate), and FRC (Family Registration Certificate): what documents are needed, who is eligible, how to apply, and how to track an application. The answers exist, but they are spread across 26 official PDF guides that most people never read. The goal is an assistant that answers these questions instantly, in plain language, using only official NADRA documents.

2. The Approach: Retrieval-Augmented Generation (RAG)

Instead of training or fine-tuning a model (expensive, slow, and hard to keep updated), the system uses RAG: the official PDFs are converted into a searchable knowledge base; for every question, the most relevant passages are retrieved and handed to a large language model, which must answer strictly from those passages. This keeps answers grounded in official text, makes updates as simple as re-running ingestion when a PDF changes, and — critically for a government context — lets the system refuse to answer when the documents do not contain the answer, instead of guessing.

The pipeline has three stages:

- Ingestion (one-time): extract text from the 26 PDFs, split it into ~1,000-character chunks, tag each chunk with its source document, page, and service category, and store everything in a local vector database.
- Retrieval + answering (per question): embed the user's question, fetch the top 5 most similar chunks, and pass them to the LLM with strict grounding rules and mandatory source citations.
- Interfaces: a terminal chat for development and a Streamlit web page for the demo.

3. Technology Choices and Why

A hard constraint shaped every choice: the entire stack must be free — no paid APIs, no credit card, no servers to rent.

Component	Choice	Why
Framework	LangChain (Python)	Prebuilt PDF loaders, text

		splitters, and vector-store integrations; avoids rewriting plumbing.
PDF extraction	pypdf	Free and fully local; standard starting point for text-based PDFs.
Embeddings	sentence-transformers/ all-MiniLM-L6-v2	Runs locally on CPU with no API key or usage limits; the standard baseline for English semantic search; small (22M parameters) so ingestion and queries stay fast.
Vector database	ChromaDB	Free, embedded, persists to a local folder; no server to install or maintain.
LLM	Llama 3.3 70B on Groq (free tier)	A 70B-class model strong enough to follow strict grounding rules; Groq's free tier offers fast inference and generous daily limits with no credit card — the project's only external dependency is one free API key.
Demo UI	Streamlit	A working chat page in ~50 lines; free and local.

4. Grounding Rules (the Government-Context Requirement)

- Answer only from the retrieved passages — never from the model's own knowledge.
- If the documents do not contain the answer, say so explicitly and point the citizen to the NADRA helpline (1777) and www.nadra.gov.pk — never guess.
- State fees, timelines, and document lists exactly as written in the source.
- Every answer cites the source document and page.

5. Phase 1 Milestones

- M1 — Knowledge base: extract, chunk, embed, and store all 26 PDFs with metadata.
- M2 — RAG pipeline: retriever + grounded prompt + Groq LLM, with a CLI for testing.
- M3 — Evaluation: a gold set of 40–60 questions across all five services, plus trap questions the system must refuse; measure retrieval hit rate, answer correctness (LLM-as-judge plus manual review), and refusal accuracy; tune chunking, top-k, and the prompt based on results.
- M4 — Demo and documentation: Streamlit chat page and a README with setup, architecture, and evaluation results.

6. Known Risks

- Some PDFs are bilingual (English/Urdu) or scanned images — text extraction quality must be verified early, with OCR or transcription as a fallback.
- Groq free-tier rate limits could slow batch evaluation; a small delay between calls handles this.
- A strict grounding prompt can over-refuse; evaluation (M3) will measure and tune this.

7. Further Steps (Beyond Phase 1)

7.1 Urdu and code-switching support

Most citizens ask questions in Urdu or in mixed Urdu-English ("CNIC renew karne ke liye kya documents chahiye?"). The plan is cross-lingual retrieval rather than building a separate Urdu knowledge base: swap the embedding model for a multilingual one (e.g. multilingual-e5-small — free and local) so questions in Urdu, English, or Roman-Urdu code-switching all retrieve the same English passages, and instruct the LLM to reply in the language the user asked in. This works because the Urdu content in NADRA's guides mirrors the English, so one clean English knowledge base can serve both languages. No new documents, OCR, or translation pipeline is required.

7.2 Voice agent

A natural next interface for accessibility — many users prefer speaking to typing. The pipeline: speech-to-text (Whisper, free — locally or via Groq's hosted whisper-large-v3, which handles Urdu and code-switched speech well), feed the transcript through the same RAG pipeline, then text-to-speech for the reply. The RAG core stays untouched; voice is a thin layer on top. A phone-line or WhatsApp voice-note front end can follow once the web version proves itself.

7.3 Other candidates

- OCR / vision-model transcription for scanned or image-only documents.
- Evaluation-driven retrieval tuning (hybrid keyword + vector search, reranking) if M3 shows retrieval misses.
- An official fee-schedule document, if fee questions prove common — the current guides describe procedures, not prices.

8. Summary

Phase 1 delivers a free, fully local-plus-one-API-key FAQ assistant that answers NADRA service questions from official documents only, refuses what it cannot verify, cites its sources, and is measured against a gold test set before anything ships. The same core then extends naturally to Urdu, code-switched text, and voice.